

Math 421: Homework 3

Solutions

Due: Sept 26, 2025

Problem 1: Prove if x and y are real numbers, then:

A. $|x - y| \leq |x| + |y|$

Proof. Let x and y be real numbers, then

$$|x - y| = |x + (-y)| \leq |x| + |-y| = |x| + |y|$$

by the triangle inequality. □

B. $|x| - |y| \leq |x - y|$

Proof. Let x and y be real numbers, then

$$|x| = |x - y + y| \leq |x - y| + |y| \Rightarrow |x| - |y| \leq |x - y|.$$

□

Problem 2: (Ross Chapter 1 3.6)

(a) Prove $|a + b + c| \leq |a| + |b| + |c|$ for all $a, b, c \in \mathbb{R}$.

Proof. Let $a, b, c \in \mathbb{R}$, then $|a + b + c| \leq |a + b| + |c| \leq |a| + |b| + |c|$, by the triangle inequality. □

(b) Let P_n be the assertion

$$|a_1 + a_2 + \cdots + a_n| \leq |a_1| + |a_2| + \cdots + |a_n|.$$

Base Case: P_1 is true because $|a_1| = |a_1|$.

Inductive Step: Assume P_n is true. Then

$$|a_1 + a_2 + \cdots + a_n + a_{n+1}| \leq |a_1 + a_2 + \cdots + a_n| + |a_{n+1}| \leq |a_1| + |a_2| + \cdots + |a_n| + |a_{n+1}|.$$

Therefore P_{n+1} holds whenever P_n holds, and the assertions holds for all $n \in \mathbb{N}$ by the principle of mathematical induction.

Problem 3: (Ross Chapter 1 3.8) Let $a, b \in \mathbb{R}$. Show if $a \leq b_1$ for every $b_1 > b$, then $a \leq b$.

Proof. We prove the contrapositive: If $a > b$, then there exists $b_1 > b$ such that $a > b_1$. Let $a > b$, then $b - a > 0 \Rightarrow \frac{b-a}{2} > 0$. Let $b_1 = b + \frac{b-a}{2}$, then $b < b_1 < a$. □

Problem 4: Prove the following theorem.

Theorem. Suppose n is an integer. If n^2 is divisible by 3, then n is divisible by 3.

Proof. We will prove the contrapositive: If n is not divisible by 3, then n^2 is not divisible by 3. Note that if n is not divisible by 3, then $n = 3k + 1$ or $n = 3k + 2$ for some integer k .

Case 1: Let $n = 3k + 1$, for some $k \in \mathbb{Z}$. Then

$$n^2 = (3k + 1)^2 = 9k^2 + 6k + 1 = 3(3k^2 + 2k) + 1$$

and is not divisible by 3.

Case 2: Let $n = 3k + 2$ for some $k \in \mathbb{Z}$. Then

$$n^2 = (3k + 2)^2 = 9k^2 + 12k + 4 = 3(3k^2 + 4k + 1) + 1$$

and is not divisible by 3.

Thus we have proven the contrapositive is true, so our original statement, if n^2 is divisible by 3, then n is divisible by 3 is also true. $\square$

Problem 5:

Theorem. $\sqrt{3}$ is irrational.

Proof. Assume, by way of contradiction, that $\sqrt{3}$ is a rational number. Then there exist $p, q \in \mathbb{Z}$, $q \neq 0$, such that $\sqrt{3} = \frac{p}{q}$. By removing common factors, we can assume that p and q are not both divisible by 3. We have

$$\sqrt{3} = \frac{p}{q} \Rightarrow 3 = \frac{p^2}{q^2} \Rightarrow 3q^2 = p^2$$

so p^2 is divisible by 3. By problem 4, this means p is divisible by 3, so there exists $k \in \mathbb{Z}$ such that $p = 3k$. Thus

$$3q^2 = p^2 = (3k)^2 = 3(3k^2) \Rightarrow q^2 = 3k^2$$

and so q^2 is divisible by 3 and q is therefore divisible by 3. This contradicts our assumption that p and q are not both divisible by 3. Thus our assumption that $\sqrt{3}$ was rational, must be false and $\sqrt{3}$ is irrational. $\square$

Problem 6:

Theorem. Prove that $\sqrt{2} + \sqrt{3}$ and $\sqrt{2} - \sqrt{3}$ are both irrational numbers.

Proof. Assume, by way of contradiction that $\sqrt{2} + \sqrt{3}$ and $\sqrt{2} - \sqrt{3}$ are not both irrational numbers. Then, at least one must be rational, if $\sqrt{2} + \sqrt{3}$ is rational, then, there exist $p, q \in \mathbb{Z}$, $q \neq 0$, such that $\sqrt{2} + \sqrt{3} = \frac{p}{q}$. Thus

$$\begin{aligned} \sqrt{2} + \sqrt{3} &= \frac{p}{q} \\ (\sqrt{2} + \sqrt{3})(\sqrt{2} - \sqrt{3}) &= \frac{p}{q}(\sqrt{2} - \sqrt{3}) \\ -1 &= \frac{p}{q}(\sqrt{2} - \sqrt{3}) \\ \frac{-q}{p} &= \sqrt{2} - \sqrt{3}. \end{aligned}$$

So that $\sqrt{2} - \sqrt{3}$ is also a rational number. Thus if one is rational, the other must also be rational. If we consider their sum,

$$\begin{aligned} (\sqrt{2} + \sqrt{3}) + (\sqrt{2} - \sqrt{3}) &= \frac{p}{q} - \frac{q}{p} \\ 2\sqrt{2} &= \frac{p^2 - q^2}{pq} \\ \sqrt{2} &= \frac{p^2 - q^2}{2pq}. \end{aligned}$$

we find that $\sqrt{2}$ is a rational number, but this is a contradiction as we have proven $\sqrt{2}$ is irrational. Therefore our original assumption must be false and both of $\sqrt{2} + \sqrt{3}$ and $\sqrt{2} - \sqrt{3}$ must be irrational. $\square$